

Nathan Hershey

Round Rock, TX | nathanhershey047@gmail.com | 937-326-3137
<https://www.linkedin.com/in/nathan-hershey-436209388>

Summary

Senior Full Stack Engineer with **8 years** of experience building cloud-native backend systems, distributed platforms, and developer-facing APIs across real estate, SaaS, and enterprise domains. Owned end-to-end delivery of financial workflows serving 500K+ users, designed multi-tenant architectures supporting 300K+ businesses, and led teams of 5 engineers on greenfield platform modules. Expert in **Python**, **Node.js**, **AWS**, and observability tooling. Works on scalable backend architecture, API security, and performance optimization for regulated workloads. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Scripting: Python, Node.js, PostgreSQL, TypeScript, JavaScript, HTML, CSS

Systems & Networking: FastAPI, GraphQL, REST APIs, AWS, CloudWatch, OAuth2, Sentry, AWS

Automation & CI/CD: CI/CD, Terraform, Terraform

Observability: Datadog

Data & Containers: MongoDB, Redis, React, Docker, Kubernetes, Postgres

Experience

Senior Full Stack Engineer | Frontdoor | Remote, US | 09/2022 - Present

- Built and maintained a real estate financial platform serving **500K+ users**, designing mortgage origination workflows with 40+ lender, title, and insurance integrations for end-to-end transaction processing.
- Migrated monolithic backend to **microservices on AWS ECS**, cutting p99 API latency from 3.2s to under 800ms and reducing infrastructure costs by 30% through container optimization.
- Designed event-driven architecture using **Redis pub/sub** for real-time status updates across 15+ services, eliminating polling overhead and improving client responsiveness by 60%.
- Established **Datadog** monitoring and alerting pipelines covering 400+ microservices, reducing mean time to detection from 12 minutes to under 90 seconds for critical production issues.
- Led team of 5 engineers on greenfield property valuation module, delivering 3-day reduction in appraisal turnaround through automated data aggregation from 8 external valuation partners.

Full Stack Developer | ZenBusiness | Remote, US | 01/2021 - 11/2022

- Developed **RESTful APIs** processing **100K+ requests daily** for business formation, registered agent, and compliance services with sub-200ms p99 latency using connection pooling and caching strategies.
- Designed multi-tenant database schema supporting **300K+ active business entities** with row-level security, ensuring strict data isolation across 50 state jurisdictions.
- Built customer-facing **React/TypeScript dashboard** with 40+ reusable components and accessible UI patterns, improving user onboarding completion by 25%.
- Integrated **Stripe payment processing** and compliance webhooks with state SOS databases, automating annual report delivery for 40K+ businesses and reducing manual processing by 70%.
- Improved CI/CD pipelines with automated testing and staged deployments, accelerating release frequency from bi-weekly to daily without downtime.

Full Stack Engineer | Northstar Software Development | TX, US | 11/2020 - 06/2021

- Developed ATS platform serving **50+ client companies**, building applicant tracking modules with custom workflow configurations and automated candidate routing rules.
- Implemented drag-and-drop kanban pipeline with **WebSocket real-time updates**, eliminating page refreshes and improving recruiter productivity by 40%.
- Built reporting and analytics features with **Charts.js** and PDF export capabilities, enabling HR teams to generate hiring reports in under 5 seconds.
- Optimized **PostgreSQL** queries and added composite indexes, reducing page load times by 40% and improving candidate search performance across datasets exceeding 100K records.

Software Engineer | Arbisoft | Remote, US | 10/2018 - 08/2020

- Built FixAlert service monitoring platform with **OAuth2** authentication and multi-channel notification delivery via email, SMS, and Slack for 200+ enterprise clients.
- Developed **REST APIs** with rate limiting and IP whitelisting for partner integrations, enabling third-party monitoring tools to consume alert data securely.
- Created alert aggregation engine with deduplication logic, reducing alert fatigue by 50% through intelligent grouping and suppression rules.
- Participated in on-call rotation for production support, triaging incidents and implementing automated remediation scripts for recurring failure patterns.

Education

Bachelor of Science in Computer Science

University of North Texas | 2014 - 2018